AAYUSH KUMAR

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Software Development Engineer | AI Backend | Game Developer

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GitHub github.com/itsAayush2004 | **Portfolio** itsaayush2004.github.io/portfolio | **Location** Jaipur, Rajasthan, India

PROFESSIONAL SUMMARY

Software Development Engineer with a B.Tech in Electronics and Communication Engineering from MNIT Jaipur (2022–2026). Founder-developer of Arthis Technologies: built and scaled ARTHIS.space to 10,000 daily active users with 70+ HTML5 games, Android and iOS apps, and a real-time multiplayer relay written from scratch in Python. At Clymora, shipped FastAPI and RAG inference to 4,500+ users at 99.8% uptime. Skilled in Python, JavaScript, C#, C++, DSA, system design, and scalable LLM integration.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, C#, Java, SQL, HTML, CSS

Backend and Databases: REST API Design, JSON APIs, FastAPI (Async), Node.js, PostgreSQL, Supabase, Database Design, Row Level Security, Authentication, Data Pipelines, Third-Party API Integration, WebSocket, TCP

AI and Machine Learning: LLM Integration, RAG Pipelines, Vector Databases, Fine-Tuning, Transformer Models, NLP, Prompt Engineering, AI Agents, Model Context Protocol (MCP), Ollama, ComfyUI, n8n

Frontend and Mobile: React, Vite, React Native, Expo, Android, iOS, Three.js, WebGL, HTML5 Canvas, Responsive Design

Game Development: Unity, C#, Unreal Engine, Mobile Game Development, Casual Games, Gameplay Programming, Game Design, Rapid Prototyping, Procedural Generation, Physics, Real-Time Multiplayer Networking, WebRTC, PeerJS, Blender Python API

DevOps and Tools: Git, GitHub, GitHub Actions, CI/CD, Linux, Docker, Cloudflare Workers, Web Security, Performance Optimization, Technical Documentation, Agile

CS Fundamentals: Data Structures and Algorithms, Object-Oriented Programming (OOP), Design Patterns, System Design, Full-Stack Development, SDLC, Software Testing, Debugging, Multithreading, Concurrency, Socket Programming, Problem Solving

WORK EXPERIENCE

Founder and Software Developer | **Arthis Technologies (Self-employed), Jaipur** — **Jun 2026 – Present**

arthis.space ↗ | arthis.land ↗ | Relay code ↗ | Space repo ↗ | Land repo ↗

- Built and scaled ARTHIS.space, a swipeable feed of 70+ HTML5 single and multiplayer games (Three.js, 2D Canvas, WebRTC), from 1 to 10,000 daily active users on Supabase and PostgreSQL.
- Wrote a zero-dependency real-time multiplayer relay (TCP and WebSocket) in pure Python stdlib, hosting 2–4 player rooms for Unity clients.
- Shipped Android and iOS apps in React Native (Expo) with a live game-preview feed and AdMob ads, tuned for low-end devices.
- Engineered a zero-cost 5-layer code-protection pipeline (obfuscation, AES-GCM encryption, Cloudflare Worker keys, WebAssembly VM) securing 78 games.
- Built Arthis.Land, a React and Vite metaverse whose Unity WebGL viewer loads .glb rooms at runtime; automated CI/CD with GitHub Actions and wrote its architecture and security documentation.
- Built internal tools with AI, including a multi-agent gameplay generator (build, score, fix loop) on Claude and local Ollama models; added RAG search and LLM fine-tuning.

Backend Developer Intern (AI) | **Clymora, Jaipur** — **Jan 2026 – Jun 2026**

- Designed and shipped 6 production REST API endpoints in async FastAPI, serving LLM inference to 4,500+ users.
- Cut API latency by 40% by rebuilding data pipelines and adding concurrent request handling, sustaining 9,000 peak daily requests at 99.8% uptime.
- Built RAG pipelines on vector databases to ground LLM answers in internal data, and trained a GPT-2 language model from scratch in Python.
- Tested, deployed, and monitored AI services on Linux within a 4-engineer Agile team.

Freelance 3D Animator and Tools Developer | **AKverse, Jaipur** — **Dec 2024 – Jan 2026**

YouTube ↗

- Developed 28 custom Blender Python add-ons (Rhubarb lip-sync, camera rigs, IK posing, render automation), replacing frame-by-frame manual animation.
- Founded AKVerse (21,000+ YouTube subscribers) and automated its Shorts production with Claude API, edge-tts, Blender, and FFmpeg; published 3D asset packs on Sketchfab.

PROJECTS

- **3D Portfolio and TravelAtlas (Three.js, WebGL):** Built a scroll-driven 3D gallery portfolio that doubles as a CV, and an interactive 3D globe charting 30+ travelled places. Portfolio ↗ | Code ↗ | TravelAtlas ↗
- **HexaBed (Unity, C#):** Procedural hex-grid terrain generator with GLB asset placement that removes manual level layout.

EDUCATION

B.Tech, Electronics and Communication Engineering | **Malaviya National Institute of Technology (MNIT, NIT Jaipur)** — **2022 – 2026**

JEE Advanced: AIR 8956 | JEE Main: 98.9 percentile | Class XII: 86.7%, Jayshree Periwal High School (2022) | Class X: 90.4%, Step By Step High School, Jaipur